

7.2 Digging in on Desalination

Read about desalination as a strategy. In the table provided, identify the advantages and disadvantages of desalination and then answer the questions.

Desalination plants: Turning seawater into drinking water

A desalination plant is a facility that removes salt and other minerals from seawater to make it safe to drink. These plants provide a reliable source of water for cities and towns where rivers, lakes, or groundwater are limited or unreliable. They are especially important in areas experiencing drought or rapid population growth.

Seawater is first filtered to remove large particles like sand or debris. Then, processes such as reverse osmosis or distillation remove salt and impurities. The clean water is sent to homes, farms, and businesses, while the leftover salty water, called brine, is carefully returned to the ocean to reduce environmental impacts.

Desalination plants are valuable because they provide a constant supply of water and reduce pressure on natural sources. However, they are expensive to build and operate, use large amounts of energy, and produce brine that can harm marine ecosystems if not managed carefully. Water from desalination may also cost more for consumers.

Overall, desalination plants are a powerful solution to water scarcity, offering reliable water even in dry conditions, but they must be carefully managed to minimise costs and environmental effects.



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1. How does a desalination plant turn seawater into drinking water?
2. Why are desalination plants important in areas with limited natural water sources?
3. What are some advantages and disadvantages of using desalination?

Advantages	Disadvantages

4. Do you think desalination is a good, long term solution to water scarcity? Justify your thinking.